Read Me File for:
Managers and Productivity in Retail

Robert D. Metcalfe, Alexandre B. Sollaci, and Chad Syverson
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Overview

The Replication folder contains 4 sub-folders that include all of the required data and codes to replicate the paper. The codes are separated in the "CodeA" and "CodeB" folders, which perform the analysis for companies A and B, respectively. The "Data" folder contains the two source datasets described below; all auxiliary datasets will be saved here as well. Finally, the "Results" folder contains another two sub-folders: "Figures", where all figures are saved, and "Tables", where data required to construct the tables in the paper are saved.

The code in this replication package constructs the analysis from the data of two large retailers using R, Stata, and Python. Within the "CodeA" and "CodeB" folders, three main files generate all of the empirical results in the paper. Simulated results are run through the files in "CodeA/NAM Simulations". The replicator should expect the code to run for about 1 hour for each company.

Data Availability

Data from the two retailers was provided under the condition that companies remain anonymous. As such, the data used for this paper cannot be made publicly available. The JPE Data Editor has conducted plausibility checks concerning data provenance under an NDA with the authors.

The data should be preserved in the restricted-access location of the Journal of Political Economy for as long as necessary for the successful replication of the paper. It should be deleted after the replication is confirmed.

The authors commit to preserving the data and code for a period of at least five years following the publication of the paper and to provide reasonable assistance to requests for clarification and replication.

Statement of Rights

I certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Dataset List

The main data sources for the analysis are the data provided by each company. They are saved on the "Data" folder as Stata data files: "A_dataset.dta" and "B_dataset.dta". A full data dictionary with the variables included in each of these datasets is provided at the end of this document in Tables 2 and 3 at the end of this file.

Several other datasets are created for the remainder of the analysis. These are listed and described in the table below. Please note that we use the letter "X" do denote datasets that are available for both companies (e.g., "X_dataset.dta" indicates that the datasets "A_dataset.dta" and "B_dataset.dta" exist for companies A and B, respectively).

Table 1: Datasets Used in Paper

Dataset	Origin	Description
X_dataset.dta	Source data	Dataset obtained from retail companies (cleaned)
X_connected.dta	Constructed; see file named: X_estimate_fe.R	Identifies all connected sets in data
X_manager_fe.dta	Constructed; see file named: X_estimate_fe.R	Contains the initial manager fixed effect estimates
X_store_fe.dta	Constructed; see file named: X_estimate_fe.R	Contains the initial store fixed effect estimates
X_manager_gid.dta	Constructed; see file named: X_estimate_gfe.R	Contains the manager group id
X_manager_gfe.dta	Constructed; see file named: X_estimate_gfe.R	Contains the manager grouped fixed effect estimates
X_store_gid.dta	Constructed; see file named: X_estimate_gfe.R	Contains the store group id
X_store_gfe.dta	Constructed; see file named: X_estimate_gfe.R	Contains the store grouped fixed effect estimates
X_dataset_fe_eb.dta	Constructed; see file named: X_fixed_effects.do	Matches the manager and store fixed effects to the source data and computes the EB-adjusted fixed effect estimates
X_dataset_gfe.dta	Constructed; see file named: X_grouped_fixed_effects.do	Matches the manager and store grouped fixed effects to the source data
X_event_study.dta	Constructed; see file named: X_event_study.do	Prepares the "X_dataset_fe_eb.dta" to run the event studies
simulate_match.dta	Constructed; see file named: selection_based_NAM.do	Keeps track of simulated manager-store matches; not used for empirical results

Note: None of the datasets can be made publicly available. See below for a description of the code scripts.

Computational Requirements

There are no specific hardware requirements to run the codes provided, and most modern laptops should be able to complete them in about 1 hour (for each company). For the paper, codes were run on a Lenovo ThinkPad T14s laptop, with 32GB of installed RAM and a 2.00 GHz processor (AMD Ryzen AI 7 PRO 350 w/ Radeon 860M). The operating system was Windows 11 Enterprise. The replicator should be able to run scripts on Stata, R, and Python. The specific libraries and dependencies required in each software are listed below.

- R (version 4.4.0) will require the following packages to be installed:
  - haven
  - rstudioapi
  - doBy
  - lfe

  - Matrix (this is not explicitly called as a library, but is required for ‘lfe’ to work properly)
- Stata (version 19) will require the following packages to be installed (all available through ssc):
  - reghdfe
  - ftools
  - did_imputation
  - event_plot
  - binscatter
  - grc1leg2
  - white_tableau scheme (can be installed through: ssc install schemepack)
- Python (version 3.11.11) will require the following libraries:
  - numpy
  - pandas
  - scipy
  - matplotlib
  - os

In addition to those, replication will also require the **ebayes.ado** and **fese_fast.ado** programs written by Adam Sacarny (see https://sacarny.com/programs/). For convenience, these can be found within the "Stata codes" sub-folder in the "CodeA" and "CodeB" folders.

Instructions to Replicators

Each "CodeX" folder contains (for X = A, B):

- A "Stata codes" folder
- Three main scripts
  - X_estimate_fe.R
  - X_estimate_gfe.R
  - run_stata_codes.do

The "CodeA" folder also contains a "NAM Simulations" folder, which has the scripts and data required to run the simulations for the NAM results:

- selection_based_NAM.do
- simulate_selection.py
- calibrate_NAM.py
- NAM_data.xlsx

Running the Replication Codes

To successfully replicate the paper, the user should

1. Run the two scripts in R ("X_estimate_fe.R" and "X_estimate_gfe.R", X = A,B) to estimate the manager and store fixed effects and grouped fixed effects.
2. Run the master file "run_stata_codes.do". This will call on several other files contained in the "Stata codes" folder and will provide all of the remaining empirical results discussed in the paper.
3. Run the scripts within the "CodeA/NAM Simulations" (no particular order required). This will provide the remaining results pertaining to the simulation exercises.

Description of the Code Scripts

We provide a detailed description of all scripts and the results they provide below. As before, we use X as a placeholder for A or B in the names of scripts that are used for both companies A and B.

- **X_estimate_fe.R** provides the initial estimates of the manager, store, and time fixed effects from equation 1 in the paper. It also adjusts the variance of the fixed effects for limited mobility bias and computes the variance/covariance shares in the four largest connected sets, as described in Table 1.

  Constructing Table 1: This script will save the variance/covariance shares in the files "j-X_cs_covar.txt", for j = 1,...,4 representing the four largest connected sets (for example, "2-A_cs_covar.txt" has the variance/covariance shares for the second largest connected set in Company A). These files can be found the "Results/Tables" folder.

  Notes:

  (a) The number of observations within each connected set shown in Table 1 is obtained after merging the "X_store_fe.dta" and "X_manager_fe.dta" datasets back into the original "X_dataset.dta", and is done in the "X_fixed_effects.do" file (it will be displayed in the Stata window when run).

  (b) The estimation of the fixed effects involves a random number generator; a seed has been set on line 17.

- **X_estimate_gfe.R** clusters all managers and stores into groups based on observables and estimates the fixed effect of each group using equation 1. As above, it also adjusts the variance and covariance of estimates for limited mobility bias and saves them in the file "X_covar_group.txt" (in this case there is a single connected set). This file is again found in the "Results/Tables" folder.

  Note:

  The estimation of the fixed effects involves a random number generator; a seed has been set on line 22.

A comment on navigating the code files: The order of the results in the code files do not always follow the order they are presented in the paper. To facilitate the navigation between code and results, the "run_stata_code.do" file indicates which results are produced in each do file it calls. These will appear as a commented code right before the "do [file name]" command.

For example, the code excerpt below indicates that the file "A_summary_stats.do" produces Figures A.1 and C.1, and Tables B.1, C.1, and D.1.

```
// Figures A.1 and C.1; Tables B.1, C.1, and D.1
do "$code_dir/A_summary_stats.do"
```

In addition, the section of code that produces each result is also identified with commented code. For example, within the "A_summary_stats.do" file, Figure A.1 is produced between lines 116 and 158. Line 115 is therefore a comment: *** Figure A.1: plot pre-move trends ***. A similar pattern holds for all Figures and Tables produced in by files below.

- **X_fixed_effects.do** [in Stata codes folder] takes the initial estimates of the fixed effects (above) and adjusts them using the Empirical Bayes Adjustment. This is done with the help of the programs **ebayes.ado** and **fese_fast.ado**.

  This script also

  - Calculates the number of managers, stores, and overall number of observations in each connected set. These resuts are used in Table 1 and discussed in Section 3.2 (see the paragraph that starts with "Our data contains about [...]").
  - Plots the distribution of fixed effects (Figure 3, Figure A.2, and Figure A.3) and the number of stores and managers in each connected set (Figure A.4);
  - Runs the out-of-sample test (Figure 4);
  - Performs the series of robustness checks discussed in Appendix C.2. This includes Figure C.2 and Figure C.3, as well as the correlations (paragraph on local shocks) and the R^2 numbers (paragraphs on persistence and learning) in Appendix C.2.
  - Estimates the full covariance matrix for manager and store fixed effects and performs the exercises discussed in the "Correlated Measurement Error" section of Appendix E.2 (see the last paragraph for the calculated ratios).

- **X_grouped_fixed_effects.do** [in Stata codes folder] analyzes the grouped fixed effects estimates, producing the Figures in Appendix E.1 (Figure E.1, Figure E.2, and Figure E.3).

- **X_event_study.do** [in Stata codes folder] runs all of the analysis related to the event studies. This includes Figure 5 and Figure 8 for Company A, Figure 6 and Figure 9 company B, and all Figures in Appendix G.

  Using the estimates from the event studies, this script also produces the results required to construct Table 4, which are displayed in Stata's interface when the code is run. The replicator can find this information next to the headers "### Table 4 (Panel A) ###" and "### Table 4 (Panel B) ###". These will indicate results before and after the change in manager by the indicator "post_change" (=0 for before change; =1 for after change). The point estimates named "b_tb" for a top-to-bottom change and "b_bt" for a bottom-to-top change.

  In addition, the script analyzes what changes for managers and stores when a store switches managers, producing Figure 7 and Figure A.5. The tenure gap mentioned in Section 4.1 are also calculated here, and will be displayed in the Stata interface under the heaer "Average tenure change (section 4.1)".

  Finally, after breaking managers and stores into terciles based on their qualities and once again looking at the initial matches and moves, the script produces Figure D.1, as well as Table 2 and Table 3. Data for both tables are saved in the "Results/Tables" folder (under "Table_2_X.dta" and "Table_3_X.dta").

- **X_summary_stats.do** [in Stata codes folder] calculates the summary statistics mentioned in the beginning of section 2 in the paper. These will be displayed in the Stata interface under the heading "number of managers per store".

  The script also does some of the preliminary analyzes of the difference between "movers" and "non-movers", plotting Figure A.1 and producing all the necessary numbers for Table B.1 (displayed in Stata's interface under the heading "### Table B.1 ###").

  Next, it estimates how different variables impact the probability that a manager moves between stores, running the regressions reported in Table D.1 (displayed in the Stata interface under the header "### Table D.1 ###").

Finally, this script compares different measures of productivity, running all of the required analysis for the discussion in Appendix C.1 (including Figure C.1 and Table C.1, displayed in Stata's interface under the header "### Table C.1 ###")

- **X__counterfactuals.do** [in Stata codes folder] does all of the analysis reported in Appendix F, including Table F.1 (again displayed in Stata's interface under the header "### Table F.1 ###").
- **B__gender__moves.do** [in Stata codes folder] (only available for company B) runs the regressions required for Table B.2 (displayed in the Stata interface under the header "### Table B.2 ###"). It also calculates the share of male and female movers (displayed in the Stata interface under the header "Prob(mover|female)") to arrive at the 7 percentage point difference mentioned in Footnote 7.

Within the "CodeA/NAM Simulations" folder the replicator will find the following files:

- **selection__based__NAM.do** is a do file that generates Figure 2 and Figure D.2.

  Note:

  The simulation requires random draws from a normal distribution; a seed has been set in line 13.

- **simulate__selection.py** is a python script that generalizes the simulations from the "selection_based_NAM.do" file into 20 rounds of (re)matching of managers and stores. This is used to produce the three charts in Figure D.3.

  Note:

  The simulation requires random draws from a normal distribution; a seed has been set in line 15.

- **calibrate__NAM.py** uses the observed variances and covariances in the data ("NAM_data.xlsx") to recover the size of the bias in the correlation between managers and stores using the expression derived in the "**Discussion and Extensions**" part of Section 3.3 and further developed in Appendix D.2.1.
- **NAM__data.xlsx** is an excel file that simply calculates the weighted averages of covariances from Table B.1. The numbers found in this file are used in the 'calibrate_NAM.py' program.

Table 2: Data Dictionary, Company A

Variable Name	Description
id	(manager,store) pair id
store_id	store id (numeric)
manager_id	manager id (numeric)
time	period id
year	year of observations
month	month of observation (name)
month_num	month of observation (numeric)
location	name of store's location
location_num	store location (numeric)
state	US State of store (name)
state_num	US State of store (numeric)
county	County of store (name)
countyfip	County FIP
statefip	State FIP
postal_code	postal code
timezone	time zone of store
remodel	indicator if store was remodeled
(comp name)startdate	date that mng started in company
positionstartdate	date that mng started position
job_description	job title
paystatus	pay status
supervisor_id	
supv_title	supervisor title
pos_start_month	month that mng started position
pos_start_year	year that mng started position
(comp name)_start_month	month that mng started in company
(comp name)_start_year	year that mng started in company
manager_tenure_pos	mng tenure in position
manager_tenure_comp	mng tenure in company
sales	total sales in month
labor_cost	cost of labor
manager_salary	manager's salary
pt_count	number of part time employees
ft_count	numer of full-time employees
oh_cost_consigned	overhead cost (consigned)
oh_cost_owned	overhead cost (owned)
oh_cost_total	overhead cost (total)
sales_area	size of sales area in store
store_area	size of store
fte_count	number of full-time equivalent employees
log_sales	log of sales
log_manager_salary	log of manager salary
log_manager_tenure_pos	log of tenure in position
log_manager_tenure_comp	log of tenure in company
log_ft_count	log of full-time employees
log_pt_count	log of part-time employees
log_fte_count	log of full-time equivalent employees
log_store_area	log of store area
log_sales_area	log of sales area

capital_cost	Cost of capital
oh_cost	overhead cost (same as oh_cost_total)
log_oh_cost	log of overhead cost
closest_competitor	distance to closest competitor (miles)
closest_[competitor 1]	distance to closest competitor 1 (company's name in brackets)
closest_[competitor 2]	distance to closest competitor 2
closest_[competitor 3]	distance to closest competitor 3
closest_[competitor 4]	distance to closest competitor 4
num_mng_2019	number of different managers in 2019
num_close_competitors	number of close competitors (within 1 mile)
sun_open_hour	hour that store opens on Sunday
sun_close_hour	hour that store closes on Sunday
sun_num_hours	number of hours open on Sunday
mon_open_hour	hour that store opens on Monday
mon_close_hour	similar as above
mon_num_hours	
tue_open_hour	
tue_close_hour	
tue_num_hours	
wed_open_hour	
wed_close_hour	
wed_num_hours	
thu_open_hour	
thu_close_hour	
thu_num_hours	
fri_open_hour	
fri_close_hour	
fri_num_hours	
sat_open_hour	
sat_close_hour	
sat_num_hours	
week_num_hours	number of hours open during week days
weekend_num_hours	number of hours open during weekends
sun_log_hours	log of hours open on Sunday
mon_log_hours	log oh hours open on Monday
tue_log_hours	similar to the above
wed_log_hours	
thu_log_hours	
fri_log_hours	
sat_log_hours	
week_log_hours	
weekend_log_hours	

Table 3: Data Dictionary: Company B

Variable Name	Description
store_id	store id (numeric)
manager_id	manager id (numeric)
time	time period
year	year of observation
month	month of observation
elec_consumption	consumption of electricity in month
elec_cost	cost of electricity in month
gas_consumption	consumption of gas in month
gas_cost	cost of gas in month
competitor_count	number of close competitors
mean_dist_km	mean distance to competitors in kilometers
mean_drivetime	mean drivetime to competitors
mean_dist_drive_km	mean driving distance to competitors in kilometers
min_distance_km	minimum distance to competitor in kilometers
min_drivetime	minimum drive time to competitor
min_dist_drive_km	minimum driving distance to competitor in kilometers
(competitor 1)	distance to (competitor 1) [name of competitor in parenthesis)
(competitor 2)	distance to (competitor 2)
(competitor 3)	distance to (competitor 3)
(competitor 4)	distance to (competitor 4)
(competitor 5)	distance to (competitor 5)
(competitor 6)	distance to (competitor 6)
(competitor 7)	distance to (competitor 7)
area_total	size of store
area_sales	size of sales area in store
store_name	store's name
format	store format (flagship, local, others)
format_num	store format (numeric)
type	store type (convenience, large, small, others)
opening_date	store opening date
region	store region (numeric)
region_name	store region name
area	store area (location, numeric)
area_name	store area name
town	store town
county	store county
postcode	store postal code
longitude	store's location longitude
latitude	store's location latitude
country	store country
location_code	store's location code
saturday_opening	Saturday opening hour
saturday_closing	Saturday closing hour
sunday_opening	similar to above
sunday_closing	
weekday_opening	
weekday_closing	
weekday_hours	number of minutes open during weekdays
saturday_hours	number of minutes open during Saturdays

sunday_hours	number of minutes open during Sundays
weekday_hours_total	total number of minutes open during weekdays
weekend_hours	number of minutes open during the full weekend
week_hours	total number of minutes open over the entire week (including weekends)
revenue_total	total revenue during month
manager_startdate	date when manager started job
months_on_job	number of months that manager has spent on job
hc_count	number of employees
fte_count	numer of full-time equivalent employees
elec_use_sqft	electricity consumption per squared foot in store
gas_use_sqft	gas consumption per squared foot in store
elec_price	electricity price (constructed as cost/consumption)
gas_price	gas price (constructed as cost/consumption)
energy_cost	energy (electricty + gas) cost
log_revenue_total	log of total revenue
log_energy_cost	log of energy cost
log_fte_count	log of full-time equivalent employees
log_hc_count	log of number of employees
location_id	store's city/location (numeric)
location_name	store's city/location name
location_type	location type (admin county, metropolitan, borough, etc.)
number_active_managers	number of active manager in store
multiple_managers	indicator of multiple managers in store
number_stores	number of store that manager in observation works in
multiple_stores	indicator of manager that works in multiple stores
open_year	year when store was opened
open_month	month when store was opened
store_age	store's age
log_store_age	log of store age
log_store_area	log of store size
log_week_hours	log of minutes open during weekdays
log_weekend_hours	log of minutes open during weekends
female	indicator of female manager
title	manager title (Mr., Mrs., Ms., others)
firstname	manager's first name
lastname	manager's last name
doj	date when manager joined company
join_year	year when manager joined company
join_month	month when manager joined company
manager_tenure	manager's tenure in job/position
manager_tenure_start	manager's tenure in company
log_manager_tenure	log of manager tenure in position
log_manager_tenure_start	log of manager tenure in company when they became a manager